

Widget Factory

Error Rate Audit

A data-driven investigation into shift performance, job mix bias,
and a classic statistical trap — prepared for Manufacturing Director Don.

09 April 2026 | Ashish Vaidya

The Situation

Don is facing a crisis — error rates are surging. He calls in the 3 shift managers.

Al

Claims to be best

*"My team has the lowest error rate!
You should fire Bob — he's the
problem."*

Bob

Confident in his team

*"My team is top notch. We have
excellent error rates."*

Carl

Silent on the matter

*Carl did not make a specific claim
about his team's performance.*

Don needs to know: who's telling the truth? And who — if anyone — should be fired?

The Raw Numbers

One month of factory data across all 3 shifts

Manager	US — Pass	US — Fail	Overseas — Pass	Overseas — Fail	Total Jobs	Total Errors
Al	394	24	50	6	474	30
Bob	140	5	846	80	1,071	85
Carl	246	12	201	20	479	32

Bob Alone handles

2.26×

more jobs than Al & Carl

Overseas error rate is

~5 %

higher than US for everyone

Al's raw overall rate

6.33%

looks best — but is it?

WHAT IS SIMPSON'S PARADOX?

When the numbers lie to your face.

A trend that appears in overall data can completely REVERSE when you look at subgroups separately.

In our factory:

AI looks best overall (6.33%)

Bob looks worst overall (7.94%)

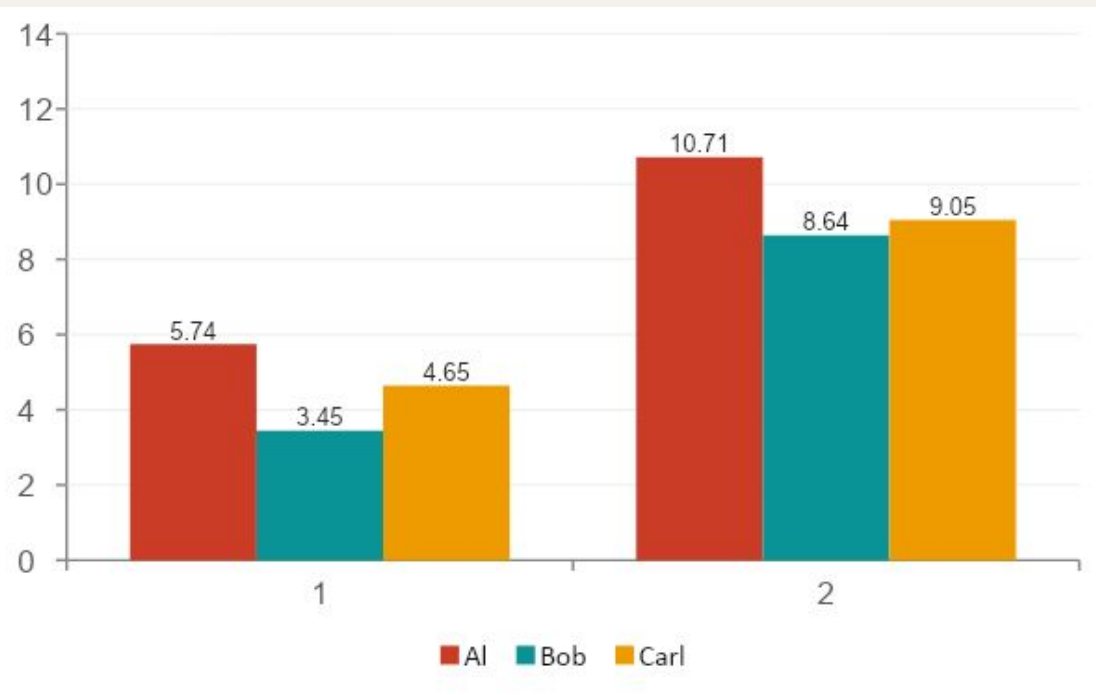
But when we compare the same type of job:

Bob beats AI on US jobs AND on Overseas jobs.

*The culprit: AI barely does overseas work (harder jobs).
Bob does most of the factory's overseas jobs.*

The Fair Comparison — Same Job Type

When we compare AI, Bob & Carl on the exact same type of job, a very different picture emerges:



KEY TAKEAWAY



Bob has the lowest error rate for US jobs — 3.45% vs AI's 5.74%



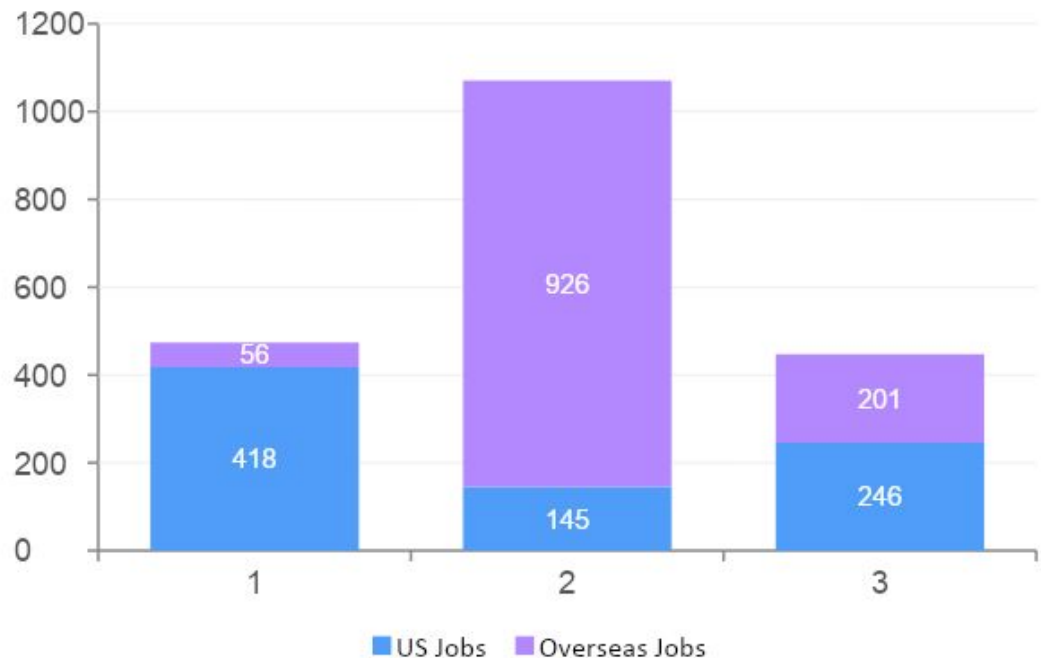
Bob has the lowest error rate for overseas jobs — 8.64% vs AI's 10.71%



AI is WORST in both categories despite claiming to be best

Why the Overall Rate is Misleading ?

Each manager handles a completely different workload. Overseas jobs are harder for everyone.



1. Al's overseas share

11.8%

Almost all US — the easier jobs

2. Bob's overseas share

86.5%

Mostly overseas — the harder jobs

Bob's total volume

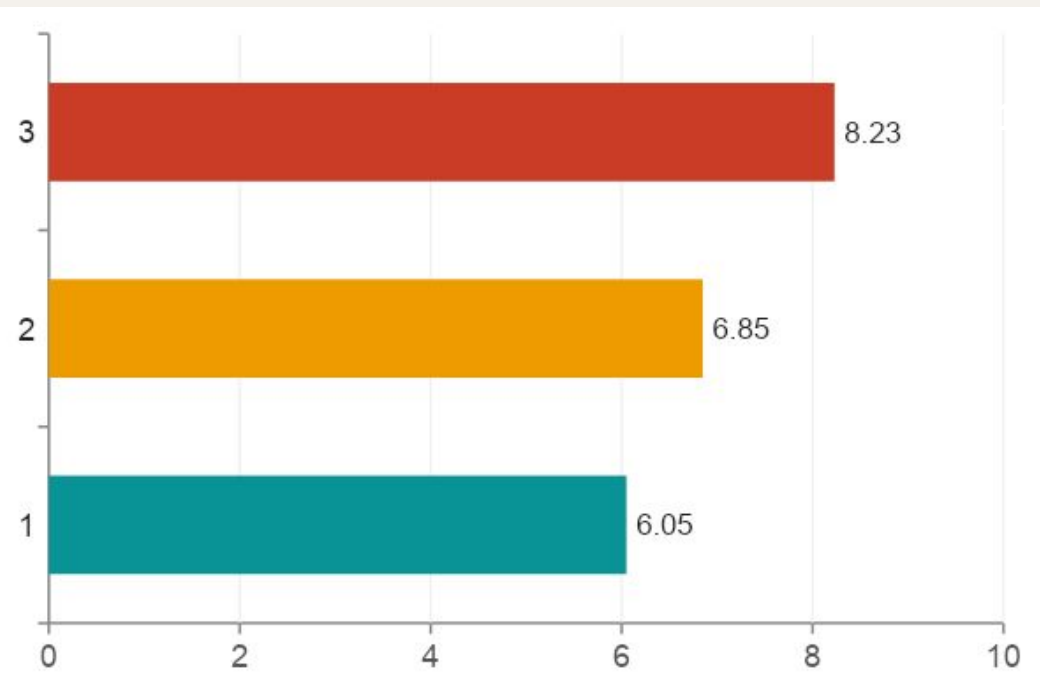
1,071

2.2× more jobs than Al (474)

Conclusion: We can't compare managers who do different types of work using the same overall metric.

The True Performance Ranking

We standardise by giving every manager the same 50% US / 50% overseas job mix — then rank them.



#1

Al

Error rate: 8.23%

Worst when job mix is held equal. Claim is false.

#2

Carl

Error rate: 6.85%

Solid, balanced performance across job types.

#3

Bob

Error rate: 6.05%

Best on US AND overseas. Deserves recognition.

Formula: $(US\ error\ \% \times 0.5) + (Overseas\ error\ \% \times 0.5)$ | Lower is better

The Bigger Problem Nobody Mentioned

Overseas jobs fail at roughly double the rate of US jobs — across ALL three managers.

US job error rate (avg)

5.05%

Across all 3 managers

+4.15
% gap

Overseas error rate (avg)

9.20%

Across all 3 managers

Possible Problems for the Overseas Error:



Different specs

Overseas widgets may have more complex or stricter requirements.



Tooling gaps

Equipment may not be optimised for overseas product variants.



Training

Staff may be less trained on overseas product requirements.



Materials

Different raw materials or components from overseas suppliers.

Key Findings

01

Al's Claim is False Al has the HIGHEST error rate on both US Jobs (5.74%) and Overseas Jobs (10.71%). His low overall rate is purely because he was assigned the easiest jobs.

02

Bob is the Best Performer

Bob has the LOWEST error rate in both product lines. His higher overall rate (7.94%) is entirely caused by handling 86% overseas jobs — the hardest category.

03

Carl is Performing Adequately

Carl's error rates are solid for both US (4.65%) and Overseas (9.05%). He handles a balanced mix and poses no concern at this stage.

04

Simpson's Paradox in Action

Raw totals mislead because job types are not distributed equally. Like-for-like comparison is the only valid approach.

Recommendations

1. **[Urgent]** *Do NOT Fire anyone (Especially Bob)* - Bob is the best-performing manager. Firing him based on raw overall rate would be a costly mistake driven by statistical illusion. He deserves recognition, not termination.
2. **[Urgent]** *Investigate Al's Performance* - Al has the worst error rate in both job types. His team needs immediate quality review in order to understand what challenges they're facing.
3. **[High]** *Learn from Bob's Best Practices* - Bob handles 54% of total factory volume with the best adjusted error rates. Document his team's process and share it across all shifts.
4. **[Medium]** *Rebalance Job Allocation* - Distribute US and Overseas jobs more evenly across all 3 shifts so comparisons become valid and every manager is assessed fairly.
5. **[Medium]** *Standardize Error Reporting* - Track US and Overseas error rates separately per shift. Replace overall rates with job-type dashboards to prevent future misdiagnosis.